

What this project computed, and how far it can be trusted

A short account of the work behind *Extremality shift of rotating black holes at finite Gauss–Bonnet coupling*: the question, the quantity that answers it, the method, the result, and — at the same length as the result — what the evidence does and does not support. The manuscript is `manuscript/main.pdf`; a longer walk-through in Spanish is `manuscript/explicacion-en-espanol.md`.

The question

A rotating black hole in Einstein gravity cannot carry arbitrary angular momentum at a given mass. For Kerr the bound is $M^2 \geq J$; in five dimensions, for the Myers–Perry solution with two equal angular momenta $J_1 = J_2 = J$, it is $M \geq \frac{3}{2}\pi^{1/3}J^{2/3}$ (see [arXiv:0801.3471](#) for the higher-dimensional solutions and their phase structure). The black hole that saturates it is extremal: its temperature vanishes and its horizon degenerates.

That bound belongs to the field equations, not to black holes in general. Add a higher-curvature term to the action and it moves. The question here is quantitative and narrow: *at fixed angular momentum, how does the extremal mass change when the coefficient α of a Gauss–Bonnet term is increased, beyond an infinitesimal perturbation of Einstein gravity?* The theory is

$$I = \frac{1}{16\pi G} \int d^5x \sqrt{-g} (R + \alpha \mathcal{L}_{\text{GB}}), \quad \mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad (1)$$

the unique quadratic invariant whose field equations stay second order, and the first nontrivial Lovelock correction in five dimensions. Varying α means comparing equilibrium solutions of theories with different couplings. It does not describe a coupling that changes with time inside one spacetime.

Two lines of work make the sign of the displacement interesting. The weak gravity conjecture reads a shift of an extremality bound as a statement about what a black hole may decay into, and the Goon–Penco relation ([arXiv:1909.05254](#)) ties the same shift to the entropy correction of nearby non-extremal states. Both are normally stated for *charged* black holes, where $M \geq M_{\text{ext}}(Q)$ compares a mass with a gauge charge. The family studied here is uncharged and carries angular momentum instead; J is not a gauge charge and none of the spectrum arguments transfers. The Goon–Penco relation is therefore used only for what it is here, an identity of the extended first law, and no weak gravity consequence is drawn from it.

What was already known

There is no closed-form rotating black hole in Einstein–Gauss–Bonnet gravity, and the literature reaches finite coupling by two routes.

Perturbatively in α . Reall and Santos ([arXiv:1901.11535](#)) showed that the $\mathcal{O}(\alpha)$ thermodynamics follows without ever solving for the corrected metric; this was done for Myers–Perry in a general quadratic-curvature theory in [arXiv:2405.04576](#), and the corrected metric in [arXiv:2512.23797](#). For the equal-spin family, Ma, Li and Lü ([arXiv:2009.00015](#)) obtained the extremal mass at first order,

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2). \quad (2)$$

The initial slope is positive and equal to π . Equation (2) says nothing about the slope once α is no longer infinitesimal, and the slope can change a great deal while keeping its sign.

Numerically. Imposing $J_1 = J_2$ enhances the symmetry enough to factor out the angular dependence: the field equations become ordinary differential equations in the radius. Brihaye, Kleihaus, Kunz and Radu ([arXiv:1010.0860](#)) used that reduction to build the non-extremal equal-spin family at finite coupling, and obtained the entropy of its extremal limit algebraically from the near-horizon geometry. Kleihaus, Kunz and Radu ([arXiv:2303.12471](#)) then constructed the extremal solutions *directly*, in a gauge that puts the degenerate horizon at the origin, and mapped the domain of existence: the Einstein bound $j \leq 1$ survives at finite coupling, the static mass bound $M > 3\pi\alpha/4$ holds for the rotating solutions too, and the extremal set is conjectured to end at the nakedly singular static configuration with $M = 3\pi\alpha/4$, $J = S = 0$.

The gap, stated carefully. It is worth being precise here, because the honest version is narrower than the tempting one. Scale invariance makes the extremal branch a single curve in the reduced variables of

[arXiv:2303.12471](#), and that curve already fixes the extremal mass: with $\mu = M_{\text{ext}}/J^{2/3}$, $y = \alpha/J^{2/3}$, and x, j the published scaled coupling and spin, the definitions alone give

$$\mu = \mu(0) j^{-2/3}, \quad y = \frac{4x\mu}{3\pi}, \quad \mu(0) = \frac{3}{2}\pi^{1/3}. \quad (3)$$

So $M_{\text{ext}}(J, \alpha)$ is contained in the published domain of existence. What is not contained there is its *derivative*. That function is not reported, and a slope read off a plotted domain boundary is not a quantity one would set beside the coefficient π of (2). Nor is differencing the masses of separately constructed extremal solutions, since each carries its own construction error and the difference is taken between neighbouring couplings. The derivative is what this project measures, and it is measured rather than differenced.

The quantity that answers it

Promoting α to a thermodynamic variable gives the extended first law of Lovelock gravity, with a conjugate potential Ψ :

$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha, \quad \boxed{\Psi = \left(\frac{\partial M}{\partial \alpha} \right)_{S, J}}. \quad (4)$$

The factor of two counts the two equal spins. “Potential” here does not mean an extra field or a function of the radius: Ψ is one number attached to each solution, measuring how its mass responds to a change of theory at fixed entropy and angular momenta, as temperature is a derivative of energy with respect to entropy.

Two facts make Ψ the right object. First, on a regular extremal branch the chain rule and (4) give $\partial M_{\text{ext}}/\partial \alpha|_J = T \partial S_{\text{ext}}/\partial \alpha|_J + \Psi$, and the first term drops as $T \rightarrow 0$ provided the extremal entropy derivative stays finite, so

$$\left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = \lim_{T \rightarrow 0} \Psi. \quad (5)$$

The slope is therefore obtained *pointwise* at each coupling, from one solution, and not reconstructed from a difference of masses. Second, the solver does not hold S and J fixed — it holds the horizon radius r_H and angular velocity Ω_H fixed — but the first law converts between the two,

$$\Psi = \left. \frac{\partial M}{\partial \alpha} \right|_{r_H, \Omega_H} - T \left. \frac{\partial S}{\partial \alpha} \right|_{r_H, \Omega_H} - 2\Omega_H \left. \frac{\partial J}{\partial \alpha} \right|_{r_H, \Omega_H}, \quad (6)$$

which subtracts from the total mass change the parts owed to entropy and spin and leaves the part owed to the change of theory.

How it was computed

The solutions. The equal-spin ansatz has four unknown radial amplitudes b, f, h, w . They are represented at Chebyshev–Lobatto nodes in a compactified radius, with the boundary behaviour factored out, and the discretised system is solved by damped Newton iteration at fixed (r_H, Ω_H, α) . The production dataset is 356 accepted solutions at 13 couplings from $\alpha = 0$ to $\alpha = 1/2$ in units $r_H = 1$; every one is a solve of the nonlinear problem at that coupling, not a perturbation about $\alpha = 0$.

The methods are those standard for stationary black holes in higher dimensions, reviewed in [arXiv:1510.02804](#).

The response, without a step size. The three derivatives in (6) could be obtained by re-solving at $\alpha \pm \Delta\alpha$ and differencing, which trades a truncation error growing like $\Delta\alpha^2$ against a cancellation error growing like the solver tolerance divided by $\Delta\alpha$ — a trade-off that worsens exactly where the solutions are hardest to converge. Instead, writing the discretised problem as $R_N(u; \alpha) = 0$ and using that it vanishes *identically* along the branch of roots,

$$A \frac{du}{d\alpha} = - \frac{\partial R_N}{\partial \alpha}, \quad A = \frac{\partial R_N}{\partial u}, \quad (7)$$

which is the implicit function theorem. One linear solve with the Jacobian the last Newton step already used returns $du/d\alpha$ exactly, with no step to refine; propagating it through the charge functionals gives Ψ . Local partial derivatives are taken by complex step. Against real central differences the response agrees at the second-order rate, the error ratios along a halving ladder falling between 3.951 and 4.028.

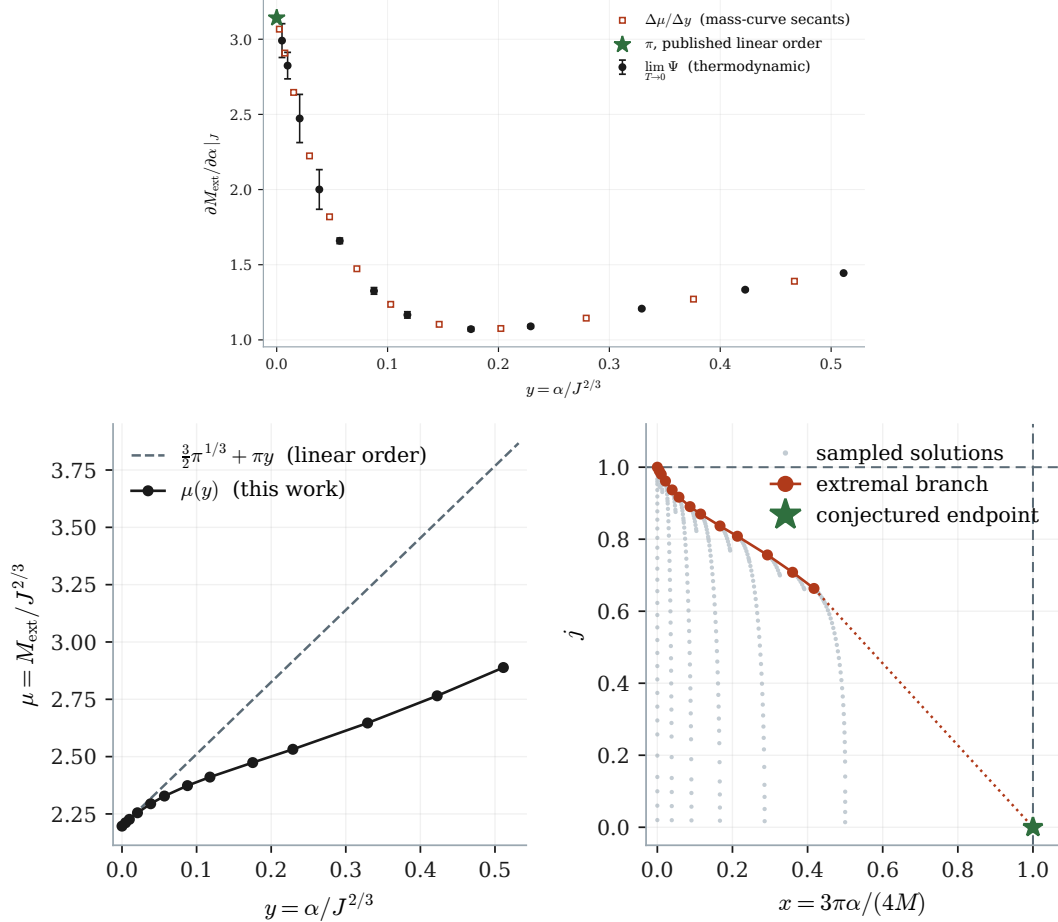


Figure 1: Top: the measured slope against $y = \alpha/J^{2/3}$, with the published linear-order value π as a star and secants of the mass curve as open squares. Bottom left: the extremal mass the slope integrates to, against the first-order line. Bottom right: the same branch in the (x, j) domain-of-existence plane of [arXiv:2303.12471](#). Circles are $T = 0$ intercepts of fits, not constructed solutions.

The extremal limit. Following each family towards $T = 0$, the coldest state reached has scaled temperature $\tau = TJ^{1/3}$ down to 3.34×10^{-6} , and the quoted extremal values are $T = 0$ intercepts of cubic fits through the six coldest states. *No solution is constructed at $T = 0$:* every extremal number is an extrapolant, and the distance each fit had to travel is reported beside it. The quoted bands are the spread of the intercept over twenty-one fitting variations — degrees one to three, six- and twelve-point windows, the coordinates T and τ , two non-polynomial forms. They measure sensitivity to stated modelling choices; they are not confidence intervals and cannot detect an error common to all.

What came out

The slope $\partial M_{\text{ext}}/\partial \alpha|_J$ is positive across the whole sampled range, so the extremal mass at fixed angular momentum increases with the coupling and the bound on the scaled spin tightens. It is not constant: it falls well below the perturbative value π , reaching 1.07 near $y = 0.175$, and then rises again to 1.44 at $y = 0.511$. Both moves exceed the fit sensitivities of the rows involved by more than an order of magnitude. Where the turning point lies *between* the sampled couplings is not resolved, which is why we quote a lowest sampled value and not a minimum.

Integrating the slope gives the mass itself: over the sampled range μ rises by 31%, while (2) runs above it and overshoots by a factor 1.32 at the largest coupling. The fall of the slope is a fall in the *rate* — the extremal mass grows more slowly than linear order predicts, not more slowly than at $\alpha = 0$.

Three ways to read it. The result becomes less abstract when restated.

1. *The spin bound, and the horizon.* By (3) a positive μ' is the same statement as a decreasing j_{ext} , so the

bound $j \leq 1$ reported in [arXiv:2303.12471](#) is the integrated form of the sign measured here pointwise. Equivalently, setting $T = 0$ in the Smarr relation gives $\alpha\Psi_{\text{ext}} = M_{\text{ext}} - 3\Omega_H J$, so a positive slope is exactly $M_{\text{ext}} > 3\Omega_H J$: the two sides coincide at $\alpha = 0$, first order says the excess opens linearly, and the measurement says it keeps opening at a rate that falls and then recovers.

2. *The geometry.* At fixed T and Ω_i the Gibbs potential is the on-shell action per unit time, and the only explicit α -dependence of that action is the Gauss–Bonnet term, so $\Psi = -(16\pi)^{-1} \int_{\Sigma} \sqrt{-g} \mathcal{L}_{\text{GB}} d^4x$ over a stationary exterior slice: a positive Ψ is a negative integrated Gauss–Bonnet density. Since the first law gives $\partial S / \partial \alpha|_{M,J} = -\Psi/T$, it is likewise a decrease of entropy at fixed mass and spin — compatible with entropy per spin along the *extremal* branch growing by a factor 4.8, because extremal states at fixed J have different masses as α changes.
3. *The turning point.* With $\omega = \Omega_H J^{1/3}$, the same Smarr relation gives $3\omega = \mu - y\mu'$ and hence $3\omega' = -y\mu''$. For $y > 0$ the inflection of the extremal mass is exactly the stationary point of ω , and the minimum of the slope is a *maximum* of ω : the extremal horizon with the largest angular velocity in units of $J^{-1/3}$. That coupling is not otherwise distinguished, and is not a boundary of the domain of existence.

Why the slope has to rise. The bound $M > 3\pi\alpha/4$ is $\mu(y) > 3\pi y/4$. If μ' stayed below some $c < 3\pi/4$ everywhere, then $\mu(y) \leq \mu(0) + cy$, which violates the bound once $y > \mu(0)/(3\pi/4 - c)$. So μ' must return to at least $3\pi/4 = 2.356$, a value above every slope measured here: the observed rise is required, not accidental. If the extremal branch also terminates as conjectured in [arXiv:2303.12471](#), the average slope tends to 2.356. The sampled range stops well short of that — the largest x reached on the extremal branch is 0.4170, against a conjectured endpoint at $x = 1$ — so this is consistency, not evidence.

How far it can be trusted

The response solve is the one step with no analytic counterpart, so it is checked against four things that do not use it.

check	what it tests	worst deviation
Boulware–Deser static limit	M, T, S, Ψ in closed form at finite α	2.60×10^{-9}
Published perturbative $\Psi_0(q)$	the response at $\alpha = 0$, at every spin	2.86×10^{-8}
Smarr relation as an equation for Ψ	charges only: no Jacobian, no linear solve	1.45×10^{-6}
On-shell Gauss–Bonnet action	curvature only: not even the charge extraction	4.67×10^{-7}
Near-horizon entropy function	S_{ext}/J at finite coupling, algebraically	4.06×10^{-6}
First law in the spin direction	internal consistency of the observables	1.35×10^{-8}
$G_{\mu\nu} + \alpha H_{\mu\nu} = 0$ off-grid	equation closure, independent evaluator	9.76×10^{-9}
Einstein extremal limits	$\mu(0)$ and $\Psi_{\text{ext}}(0)$ against $\frac{3}{2}\pi^{1/3}, \pi$	$4.56 \times 10^{-11}, 1.19 \times 10^{-7}$
Radial resolution, 26 full re-walks	the whole pipeline at two coarser N	6.89×10^{-5}

The last two rows deserve comment. The Einstein limit is recovered from the same continuation and extrapolation used at finite coupling, with the analytic constants imposed nowhere; agreement there does not establish the error at finite y . The resolution study re-ran the complete ascending sequence and the complete $T = 0$ extrapolation at two coarser base resolutions at every coupling, and the intercept moves by less than the fit sensitivity in all 26 sequences — which bounds the resolution effect but is not a convergence proof, for three reasons named in the manuscript’s appendix.

Limitations. The extrapolation is the leading one, and the bands around it are sensitivities, not error bars. The coupling coverage is finite, the spin coverage uneven, and the calculation is restricted to equal spins and to the branch continuously connected to Myers–Perry; nothing here shows that branch is unique, stable, or the lowest in mass at the same angular momenta. None of the checks closes the gap that matters most: the mass-curve comparison shares systematic errors with the potential, the near-horizon route determines entropy rather than mass, and the Smarr and on-shell routes remove the response solve from the chain but not the solutions. Finally, this is classical Einstein–Gauss–Bonnet with the action (1). In a gravitational effective field theory, terms beyond Gauss–Bonnet contribute at the same order as the higher powers of α kept here, so the non-monotonic slope is not a universal prediction of a derivative expansion.

One comparison that does not close. A separate reconstruction of the non-extremal family gives an outer ergosurface radius 1.102101 against the 1.104 quoted in [arXiv:1010.0860](#), at $\alpha = 1/2$ in our convention. It is unresolved, recorded as a declared negative result rather than absorbed into a tolerance, and not offered as evidence of an erratum. It tests the metric functions away from the horizon and touches neither the charge extraction, the response solve nor the extrapolation; the near-horizon comparison that does test the same paper’s coupling conversion agrees to 9.17×10^{-14} .

New, reproduced, and cited

New here: Ψ measured on rotating solutions at finite coupling, across the whole family and not only at its extremal end, with the sign change it undergoes; the extremal slope obtained pointwise at 13 couplings, its minimum and subsequent rise resolved; the extremal mass curve it integrates to, and the size of its departure from (2); the reading of Ψ as an on-shell Gauss–Bonnet action, tested against the response on 43 archived solutions; and the identification of the slope minimum with the maximum of $\Omega_H J^{1/3}$.

Reproduced rather than new: the equal-spin family of [arXiv:1010.0860](#) and its profiles; the closed-form static solution; the $\mathcal{O}(\alpha)$ potential implied by [arXiv:2405.04576](#) and the first-order extremal mass π of [arXiv:2009.00015](#); the near-horizon extremal entropy of [arXiv:1010.0860](#); and the extremal branch itself, which [arXiv:2303.12471](#) constructs directly and this work reaches by extrapolation, with its bounds $j \leq 1$ and $x < 1$.

Cited, not reproduced: the Myers–Perry and Boulware–Deser solutions (oracles, every closed form checked against the solver, but not rederived), Lovelock’s theorem, the Wald and Jacobson–Myers entropy formulae, Sen’s entropy function formalism, the covariant phase space derivation of the extended α -term, and the Goon–Penco universality claim — of which only the first-law identity is used, and that is rederived.

The apparatus

One discipline shaped the repository more than any other: **no number that appears in the paper is typed into the paper**. Every figure, table and quoted quantity is generated from the recorded measurement into `numbers.tex` and read back as a macro, so the PDF, the figures, the tables and the data cannot disagree: rebuilding either reproduces them or fails. This document draws on the same file.

Around that sit the checks that make a claim auditable rather than merely stated. A static checker verifies that every generated quantity is used and every used one generated, that no literal decimal survives in the prose, and that every citation resolves. Each result carries a provenance record naming its command, inputs and their hashes, audited against the bytes on disk, with the two exceptions it cannot resolve documented rather than silenced. Failed checks are declared in `artifacts/limitations.json`, and a test asserts that the declared failures still match the actual ones, so the list cannot grow into a way of muting the audit.

Two lessons are worth recording. Re-running the same boundary-value problem on a different linear-algebra stack moved an extrapolated intercept by more than either genuine resolution change at the same coupling: the walk is adaptive, so a different BLAS changes which states pass the acceptance gate and therefore which states get fitted. And the quadrature for the on-shell integral *cannot* be refined — $\mathcal{L}_{\text{GB}}$ is a cancellation decaying like r^{-8} while the compactification amplifies what survives, so past about eighty nodes the answer degrades and then diverges. In both, the obvious convergence diagnostic measures something other than what it appears to.